Introductory Economics for Business and Finance (IC103)

Tutorial 2



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Where business comes to life

Question 1

Is this statement true or false? “A price rise is likely to lead to a fall in total expenditure if the demand curve is elastic.”

Question 1

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True, because by definition the percentage change in quantity demanded is greater than the percentage change in price.

Total expenditure is:

$$TE = P \times Q$$

If demand is elastic, then PεD > 1:

$$P_{\varepsilon_D} = \frac{\%\Delta Q_D}{\%\Delta P}$$

This means quantity demanded changes by a **larger** percentage than price.

For example, suppose:

- price rises by 10%
- quantity demanded falls by 20%

The fall in quantity is proportionately larger than the rise in price, so overall expenditure falls.

Question 2

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Consider the following information on income and expenditure on certain classes of goods:

	Income (£m)	Bread and cereals	Beer	Tobacco	Air travel
2018	663	42.8	84	68.1	26.1
2019	695	43.6	85	67.6	31.6

Calculate the income elasticity of demand for each category of goods, using both the mid-point and percentage methods.

Note: Strictly speaking, income elasticity of demand should use quantity demanded.

In this exercise, the expenditure figures are being used as an approximation for demand, effectively assuming that prices have not changed significantly.

Question 2

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	Income (£m)	Bread and cereals	Beer	Tobacco	Air travel
2018	663	42.8	84	68.1	26.1
2019	695	43.6	85	67.6	31.6
midpoints	679	43.2	84.5	67.85	28.85

Using % method: Income elasticity = $\dfrac{\frac{Q_2-Q_1}{Q_1}}{\frac{Y_2-Y_1}{Y_1}}$

bread $\dfrac{\frac{43.06-42.8}{42.8}}{\frac{695-663}{663}} = .39$

Demand rises when income rises, but by a smaller percentage than income.

Question 2 continued

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Beer $\frac{\frac{85-84}{84}}{\frac{695-663}{663}} = .25$

Demand rises when income rises, but by a smaller percentage than income.

Tobacco $\frac{\frac{67.6-68.1}{68.1}}{\frac{695-663}{663}} = -.15$

Negative income elasticity means that as income rises, demand for tobacco falls slightly.

Air travel $\frac{\frac{31.6-26.1}{26.1}}{\frac{695-663}{663}} = 4.37$

Demand rises by a larger percentage than income.

Using midpoints

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Using midpoint method: Income elasticity $= \dfrac{\dfrac{Q_2 - Q_1}{(Q_1 + Q_2)/2}}{\dfrac{Y_2 - Y_1}{(Y_1 + Y_2)/2}}$

Bread $\dfrac{\dfrac{43.6-42.8}{43.2}}{\dfrac{695-663}{679}} = .39$

Beer $\dfrac{\dfrac{85-84}{84.5}}{\dfrac{695-663}{679}} = .25$

Tobacco $\dfrac{\dfrac{67.6-68.1}{67.85}}{\dfrac{695-663}{679}} = -.16$

Air travel $\dfrac{\dfrac{31.6-26.1}{28.85}}{\dfrac{695-663}{679}} = 4.05$

Question 3

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An online bookseller wants to increase its total revenue. One strategy is to offer a 10% discount on every book sold. The bookseller knows that its customers can be divided into two groups according to their likely responses to this discount. Relevant information on book sales per week is provided below:

	Volume of sales before discount	Volume of sales after the discount
Group A	1.55m	1.65m
Group B	1.5m	1.7m

a) Calculate the price elasticity of demand for both groups using the percentage method.

b) Explain how the discount will affect total revenue from each group.

c) Imagine that the bookseller can tell which group a consumer belongs to when they login, and can choose whether or not to offer the 10% discount. Which group(s) should be offered the discount?

Question 3

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a)

For Group A:

$$P_{\varepsilon_D} = \frac{\%\Delta Q_d}{\%\Delta P}$$

$$P_{\varepsilon_D} = \frac{(1.65 - 1.55)/1.55}{-0.10}$$

$P_{\varepsilon_D} = -0.65$ -> Inelastic demand

For Group B:

$$P_{\varepsilon_D} = \frac{(1.70 - 1.50)/1.50}{-0.10}$$

$P_{\varepsilon_D} = -1.33$ -> Elastic demand

b)

For Group A. The 10% price reduction causes quantity demanded to rise by only about 6.5%. So total revenue decreases.

For Group B. The 10% price reduction causes quantity demanded to rise by about 13.3%. So total revenue increases.

c)

The bookseller should offer the discount to Group B only.

This is because the percentage increase in sales is greater than the percentage fall in price. Therefore, the discount increases total revenue from this group.

Question 4

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The following table provides part of the supply schedule for laptop computers in country A:

Price of laptop	Quantity of laptops supplied
1100	12000
900	8000

a) Use the mid-point method to calculate the price elasticity of supply when the price rises from 900 to 1100.

b) Suppose that improvements in technology enable companies to produce an additional 1000 laptops at any price point. If the price moves from 900 to 1100, is the price elasticity of supply the same as, less than or more than it was in (a)?

c) Suppose a longer time period is considered, so the quantity supplied at any price is 20% higher than in the table. Looking again at a price increase from 900 to 1100, is the price elasticity of supply the same as. Less than or more than it was in (a)?

Question 4

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a)

Price midpoint is (1100+900)/2=1000

Quantity midpoint is (12000+8000)/2=10,000

Price elasticity of supply is $\frac{\frac{8000-12000}{10000}}{\frac{900-1100}{1000}}=2$

Midpoint method:

Price elasticity $= \frac{\frac{Q_2-Q_1}{(Q_1+Q_2)/2}}{\frac{P_2-P_1}{(P_1+P_2)/2}}$

b) Now the table becomes

Price of laptop	Quantity of laptops supplied
1100	13000
900	9000

The quantity midpoint is now 11,000 and the calculation becomes:

$$\frac{\frac{9000-13000}{11000}}{\frac{900-1100}{1000}} = 1.8$$

Elasticity is lower than before. Because adding 1,000 units to both quantities raises the base/midpoint but does not change the absolute change of 4,000.

4 (c)

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Now the quantities are 9600 and 14400 with a midpoint of 12000 and the calculation becomes:

$$\frac{9600 - 14400}{12000} \Big/ \frac{900 - 1100}{1000} = 2$$

Elasticity is exactly the same as in part a).
This is because both quantity levels and their difference have been multiplied by the same factor, 1.2.

Question 5

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Consider the following elasticities published in a an academic journal article for the UK:

	Income elasticity	Own price elasticity	*Cross price elasticity (quantity of product in bold with respect to price of product in italics)*		
			Beer	*Wine*	*Spirits*
Beer	0.55	-0.13		0.04	0.09
Wine	1.23	-0.37	0.15		0.22
Spirits	1.82	-0.32	0.19	0.13	

a) To what extent would a fall in wine prices affect consumption of wine, beer and spirits?

b) If spirits prices were halved, what would be the effect on consumption of spirits?

c) Could beer producers gain from an agreement to raise prices together?

Question 5

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Own-price elasticity: what happens to demand for the same good when its own price changes?
Cross-price elasticity: what happens to demand for one good when another good's price changes?
Income elasticity: what happens to demand when income changes?

a) The own-price elasticity of wine is:-0.37
That means when the price of wine falls, the quantity demanded of wine rises.
Then we look at the cross-price elasticities with wine:
Beer with respect to wine price = +0.04 and spirits with respect to wine price = +0.13.
Because both are positive, beer and spirits behave as substitutes for wine.
So, Wine consumption **increases.** Beer consumption **falls slightly.** Spirits consumption **falls.**

b) The price of spirits is halved

$$\%\Delta P = -50\%$$

The own-price elasticity of spirits is:

$$P_{\varepsilon_D} = -0.32$$

$$P_{\varepsilon_D} = \frac{\%\Delta Q_d}{\%\Delta P}$$

$$\%\Delta Q_d = P_{\varepsilon_D} \times \%\Delta P$$

$$= (-0.32) \times (-50\%)$$

$$= 16\%$$

Quantity demanded of spirits rises by 16%.

c) Beer's own-price elasticity is -0.13. So beer demand is very inelastic.
That means if beer producers raise prices, quantity demanded falls only a little. Because the price increase is proportionately larger than the fall in sales, total revenue can increase.

Question 6

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If beef is a substitute for lamb and mint sauce is a complement to lamb, which of the following is correct?

An increase in price of lamb will (ceteris paribus) lead to:

(a) A decrease in the quantity of beef demanded – no, more beef will be demanded

(b) An increase in the quantity of beef demanded and an increase in the quantity of mint sauce demanded – yes to beef, no to mint sauce

(c) A decrease in the quantity of mint sauce demanded

(d) An increase in the quantity of mint sauce demanded and a decrease in the quantity of beef demanded – no, the opposite is true

Question 7

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If the price of a good falls from £27.50 to £22.50, and the quantity demanded by consumers increases from 15 to 25 units a week, then using the midpoint formula the price elasticity of demand is:

(a) -0.25

(b) -1.00

(c) -2.75

(d) -2.5

Question 7

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For mid-point method, the price elasticity should be:

$$P_{ed} = \frac{\Delta D / D_{mid}}{\Delta P / P_{mid}}$$

$$P_{ed} = \frac{(25 - 15)/20}{(22.5 - 27.5)/25}$$

$$P_{ed} = -2.5$$

Question 8

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If the price elasticity of demand for a good is -3.0, then

(a) 12% drop in price leads to a 36% increase in the quantity demanded

(b) 12% drop in price leads to a 4% increase in the quantity demanded

(c) £1,000 drop in price leads to a 3,000 unit increase in quantity demanded

(d) A £1,000 drop in price leads to a 333 unit increase in the quantity demanded

Question 8

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The absolute value for price elasticity is 3, which is bigger than 1. A small decrease in price will lead to a large increase in quantity demanded.

$$P_{ed} = \frac{\%\Delta Q_D}{\%\Delta P}$$

$$\%\Delta Q_D = \%\Delta P \times P_{ed}$$

$$\%\Delta Q_D = -12\% \times -3.0$$

$$\%\Delta Q_D = 36\%$$

Question 9

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Which of the following will have positive signs and which will have negative signs?

(a) Price elasticity of demand --negative

(b) Income elasticity of demand --positive for normal goods and negative for inferior goods

(c) Cross price elasticity of demand (with respect to a change in the price of a substitute) --positive

(d) Price elasticity of supply --positive